Auditing Satisfaction, Not Just Accuracy: Calibration and Evidence Consistency of LLM-Based Satisfaction Scoring under Synthetic Call-Log Shift

Preliminary research report

Project `callsat-audit` — prepared for the ECIS 2027 Track 10 theme
"Design and management of services in the age of AI"

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Abstract

Contact centres increasingly use large language models (LLMs) to infer customer satisfaction from call transcripts, often on synthetic logs because real conversations are sensitive and scarce. A satisfaction score, however, is a measurement instrument: it must be calibrated, traceable to the evidence that produced it, and stable when the data-generating process shifts. We report a preliminary stress test of a call-eval-style measurement chain—transcript, cited evidence spans, typed facts, deterministic score, survey calibration—on 420 synthetic source calls and 100-call paired targets for four shifts (disfluency, emotional arc, lexical paraphrase, combined) plus an unpaired persona-mix target. On the sealed source holdout the chain reaches quadratic-weighted kappa (QWK) 0.718 against the generator's latent satisfaction and expected calibration error (ECE) 0.050, but only QWK 0.502 and ECE 0.128 against the noisy survey it was fitted to. Lexical paraphrase is catastrophic: QWK falls to 0.098, the gold evidence-turn hit rate to 0.082, and 90% conformal coverage to 0.760. A direct 3B open-model scorer holds QWK at 0.667 under the same paraphrase, but is severely overconfident (confidence ECE ≈ 0.50) and cites real-but-often-wrong turns (citation precision 0.53–0.60). The preliminary evidence suggests that the fragile component is the span/fact interface rather than the calibration layer, that ECE alone is too blunt an audit at this sample size, and that robustness and auditability trade off. We close with the next experiments needed before any claim about frontier LLMs or real callers.

Status. Work in progress, not a submission-ready paper. All data are synthetic and all numbers describe the measurement pipeline on that data, not real customers, agents, or deployments. The LLM arm is one local 3B model on 40 source calls and their paraphrase twins; it is a feasibility probe, not a model comparison. Code, notebooks, and run artefacts accompany this report.

1 Introduction

Customer satisfaction (CSAT) is a central service-operations metric. Contact centres collect it through post-call surveys, but response rates are low and non-response is informative: the customers who are most dissatisfied are often the least likely to answer. This has motivated models that predict CSAT directly from the call, either to fill in missing survey responses or to prioritise coaching and follow-up (Manderscheid and Lee, 2024). LLMs make such scoring cheap and general, and they can, in principle, attach a rationale and an evidence span to every score (call-eval maintainers, 2026; JevList, 2026). The same technologies are being used to evaluate agent performance and compliance at scale (Chincha et al., 2025).

Synthetic call logs are an attractive development substrate: they avoid privacy exposure, can be generated at arbitrary scale, and can be annotated by construction. Yet synthetic dialogue is not automatically realistic. Mirage benchmarks synthetic contact-centre transcripts against real ones across 17 metrics in four dimensions and finds measurable deficiencies in sentiment fidelity, disfluency modelling, behavioural variation, and conversational realism; prompts optimised on synthetic transcripts underperform those optimised on real transcripts on a downstream AutoQA task (Devanathan et al., 2026). A satisfaction scorer developed and calibrated on synthetic logs may therefore inherit a realism gap that is invisible in its training loss.

The call-eval architecture is a useful concrete instance of the measurement chain we study. It extracts typed facts with evidence spans (was the request resolved, how much effort did the caller expend, what did the caller say they felt), uses deterministic code to turn those facts into a 1–5 satisfaction score and a sentiment label, fits an isotonic calibration layer against post-call surveys, and evaluates on a sealed holdout with quadratic-weighted kappa (call-eval maintainers, 2026; JevList, 2026). Its design is auditable by construction: every number can be traced to transcript lines. The public repository returned 404 on 20 October 2026; we therefore replicate the architecture from the archived project guide rather than the original code or data.[1]

The open question is not whether an LLM can produce a plausible satisfaction score. It is whether the *measurement chain* remains calibrated and auditable when the synthetic data-generating process changes. Prior work gives us three relevant but separate bodies of evidence: realism diagnostics for synthetic contact-centre dialogue (Devanathan et al., 2026; Soudani et al., 2026; Zhu et al., 2026); agreement and reliability studies for LLM annotation of affect and satisfaction (Kumar et al., 2026; Krawczyk et al., 2026); and calibration-under-shift results from adjacent classification tasks, where post-hoc temperature scaling can leave substantial residual miscalibration (Guo et al., 2017; Wang et al., 2026). What is missing is a design that varies the synthetic generation process in a controlled, paired way and asks where in the evidence chain the failure occurs.

This report starts that design. We generate paired synthetic calls in which the latent satisfaction label and survey response are held fixed while the surface realisation changes, and we measure four things: discrimination (QWK), calibration (ECE, Brier, slope), selective behaviour (conformal coverage and risk–coverage), and evidence consistency (span hit, fact accuracy, quote support, paired score/label changes). We then compare the evidence-chain arm with a direct small-LLM scorer. The preliminary contributions are:

1. A *paired-shift stress-test design* for satisfaction measurement that separates label-preserving noise (disfluency), concept shift (emotional arc), lexical shift (paraphrase), covariate shift (persona mix), and a combined worst case.

2. An *evidence-chain audit* that localises failure at the span, fact, score, or calibration layer, rather than reporting a single agreement number.

3. A preliminary empirical profile showing a large lexical-shift collapse, a narrower and statistically uncertain effect for mild shifts, and a survey-versus-construct gap.

4. A negative result for presence-based runtime gating, which motivates a contradiction-based audit, and a robustness–auditability contrast between the symbolic chain and a direct 3B LLM.

The report is deliberately a *start*: the offline extractor is a transparent lower bound, the LLM arm is small, and the shifts are synthetic transformations. Section 6 states what the design can and cannot support.

[1]The guide is archived at `https://jevlist.ai/projects/call-eval`. The maintainer's reported agreement numbers are on synthetic template-generated calls with simulated annotators; they describe the artefact on that data, not real callers.

2 Related work

Synthetic contact-centre dialogue. Synthetic dialogue generation has matured from template filling to structured, multi-agent role-play. Soudani et al. (2026) survey the space and organise it around seed-data creation, utterance generation, and quality filtering. Zhu et al. (2026) generate strategy-rich customer-support conversations and build an evaluation set from rewritten real conversations; their key lesson is that synthetic data becomes more useful when it is supervised for service strategy, not merely made to look realistic. Devanathan et al. (2026) provide the critical counterweight: realism is multi-dimensional, and synthetic transcripts can be fluent while still misrepresenting sentiment arcs, disfluency, and interaction style. Our shifts operationalise exactly those dimensions, but as controlled transformations rather than as a realism benchmark.

Satisfaction and agent-performance measurement. Manderscheid and Lee (2024) frame CSAT prediction as a distribution-matching problem: predicted scores should reproduce the survey response distribution, not just minimise per-call error. Krawczyk et al. (2026) show that interpretable, structured sales-action metrics can predict satisfaction almost as well as a text-embedding baseline ($R^2 = 0.87$ vs. 0.92) on 5,200 simulated sales dialogues, which supports fact-based scoring as a transparent alternative to end-to-end regression. Chincha et al. (2025) report an LLM-based call-centre evaluation system with high accuracy on call interpretation. The call-eval artifact (call-eval maintainers, 2026; JevList, 2026) is closest to our setting: typed facts, evidence spans, deterministic scoring, isotonic calibration, and a sealed holdout, evaluated with QWK against simulated annotators. None of these studies stress-tests the chain under paired synthetic shift.

LLM judgment and annotation reliability. Kumar et al. (2026) compare experts, crowdworkers, and LLMs on empathy annotation across four frameworks and 200 real conversations. Expert agreement is high but varies by subcomponent; LLMs approach the expert benchmark and exceed crowdworkers, but the appropriate benchmark is expert agreement, not raw classification accuracy. This matters for our design: a QWK of 0.7 against a noisy survey or a single annotator is not self-evidently good or bad. It has to be read against the reliability ceiling of the labels. Our synthetic design gives us that ceiling by construction, but also makes the labels a property of the generator.

Calibration and uncertainty under shift. Post-hoc calibration is the standard remedy for overconfident classifiers (Guo et al., 2017), and isotonic regression is a common non-parametric choice (Niculescu-Mizil and Caruana, 2005). The evidence on transfer is less encouraging. Wang et al. (2026) report a "calibration cliff" in cyberbullying detection: on Civil Comments, the worst-to-best ECE ratio between compact encoders and LoRA-adapted 7B models reaches 14.9 times, and a 9.4-times residual ratio remains after temperature scaling. Their task is not CSAT, but the structural lesson transfers: a calibration layer fitted in one distribution need not remain calibrated in another. Split conformal prediction offers distribution-free coverage under exchangeability (Angelopoulos and Bates, 2021); shift violates that assumption, which makes coverage a useful *audit outcome* rather than a guarantee. We use QWK and weighted kappa (Cohen, 1968) for ordinal agreement, and ECE, Brier, and calibration slope for probabilistic calibration.

Evidence and auditability. The call-eval design binds every score to transcript lines (call-eval maintainers, 2026; JevList, 2026). Wu et al. (2026) describe an evidence-grounded customer-service agent workflow in which models "must not bypass evidence boundaries, policy rules, or human-handoff safeguards." Our audit operationalises a weaker but measurable version of that

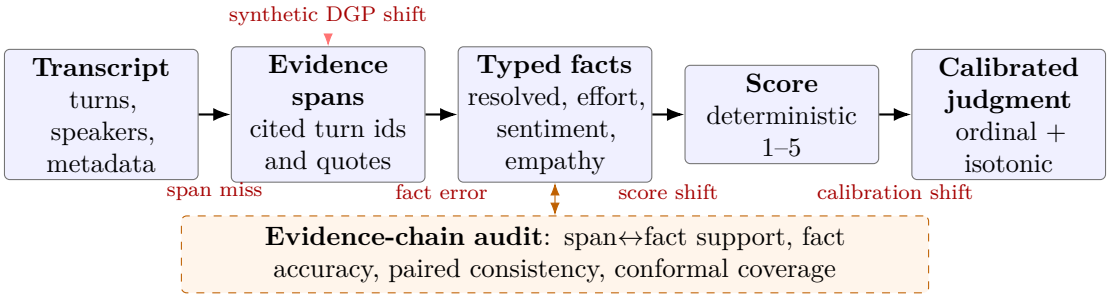


Figure 1: The measurement chain and its failure loci. The same source-fitted extractor and calibration layer are applied to each shifted target. The audit band is evaluated with both oracle-assisted diagnostics (gold evidence turns) and label-free runtime flags.

idea: a score is auditable if its cited span exists, supports the typed fact, and survives a meaning-preserving perturbation. The ECIS Track 10 call for accountability, transparency, and responsible service design (ECIS, 2027) is the organisational motivation: service managers need to know whether an AI-generated satisfaction dashboard measures satisfaction or merely fits a survey.

3 Study design

3.1 The measurement chain

Figure 1 shows the chain we study. A transcript is passed to an extractor that returns typed facts with evidence spans. A deterministic scoring function maps the facts to a 1–5 score. A calibration layer maps the score to a survey-anchored judgment. An audit layer checks the span–fact–score chain and the calibration's behaviour under shift.

3.2 Synthetic data-generating process

We generate 420 source calls: 240 train, 80 dev, and 100 sealed holdout. Each call has a persona (anxious elderly, busy professional, angry returning, calm first-timer, vulnerable financial), an issue type (billing, refund, technical, account access, cancellation), a resolution outcome (resolved, partial, unresolved), caller effort (0–3), agent empathy (0/1), and an expressed sentiment (-2 to $+2$). A structural equation combines resolution, empathy, effort, prior failures, wait time, and persona into a latent satisfaction z; cutpoints map z to a 1–5 latent label. A simulated post-call survey is a noisy draw around the latent label with about 71% response; two simulated annotators provide noisy labels for a human-ceiling check.

The transcript is rendered from templates with explicit cue sentences for each fact. To avoid a trivially perfect extractor, a fraction of cues is replaced by implicit but entailed evidence: 22% of resolution cues, 18% of empathy cues, 12% of sentiment cues, and 10% of effort cues. Gold facts are unchanged; the information is still in the transcript, but the surface pattern is gone. This creates realistic evidence scarcity without making the label unfair.

The deterministic score is a weighted sum of four normalised facts:

$$\text{raw} = 1 + 4 \cdot \text{clip}(0.55\, r + 0.20\,(1 - e/3) + 0.15\,(s+2)/4 + 0.10\, a,\ 0,\ 1)\,, \tag{1}$$

where $r \in \{0, 0.5, 1\}$ is resolution, $e \in \{0, 1, 2, 3\}$ is effort, $s \in [-2, 2]$ is sentiment, and $a \in \{0, 1\}$ is empathy. The weights are transparent and fixed before any shift is evaluated.

Table 1: Shift conditions. The first four are paired transformations of the source holdout: latent satisfaction and survey response are held fixed. Persona mix is an unpaired covariate shift with new calls.

Condition	Transformation	Intended stress
Disfluency	Fillers, repetitions, and light ASR noise inside cue spans	Label-preserving measurement noise
Emotional arc	Late frustration after resolution, or suppressed dissatisfaction	Concept shift: expressed sentiment decouples from outcome
Paraphrase	Meaning-preserving synonym replacements for cue sentences	Lexical shift: tests span robustness
Combined	Paraphrase + emotional arc + disfluency	Realistic worst case
Persona mix	Different persona weights, new calls	Covariate shift / base-rate change

All shifts are synthetic text transformations, not recordings of real disfluency, accent, code-switching, or emotional expression.

3.3 Calibration and audit layers

Calibration is fitted on source train+dev surveys only ($n = 227$). We compare two layers: an ordinal proportional-odds model on the raw score with a single temperature fitted on out-of-fold predictions, and an isotonic regression of the survey label on the raw score (Niculescu-Mizil and Caruana, 2005). The ordinal layer supplies class probabilities for ECE, Brier, selective prediction, and conformal sets; isotonic supplies point scores. The implementation uses scikit-learn for isotonic regression and the agreement metrics (Pedregosa et al., 2011). A split-conformal layer uses nonconformity $1 - p(y)$ and a 90% target coverage (Angelopoulos and Bates, 2021); the threshold is fitted on out-of-fold probabilities and applied unchanged to every target.

We report QWK (Cohen, 1968), mean absolute error, Spearman correlation, ECE

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{n} \left|\mathrm{acc}_b - \mathrm{conf}_b\right|, \tag{2}$$

Brier score, negative log-likelihood, calibration slope (OLS of observed on predicted), the large-error rate $P(|\hat{y} - y| > 1)$, and conformal coverage/set size. All estimates carry 1,000 bootstrap resamples; paired shifts use paired ΔQWK intervals. The source frame is the sealed 100-call holdout; train/dev are never included in any reported target metric.

3.4 Shift conditions

Table 1 summarises the five conditions. The first four are paired transformations of the source holdout: the latent label and survey response are held fixed, so any change in the score is attributable to the transformation. Persona mix is unpaired and changes the base rate.

3.5 Hypotheses and decision rules

We pre-specified four hypotheses. **H1** (calibration transfer): ECE increases and the calibration slope departs from one under shift. **H2** (localisation): failures occur at the span/fact layer before calibration, visible as a drop in gold evidence-turn hit rate or fact accuracy while the latent label is unchanged. **H3** (audit signal): runtime evidence flags predict large scoring errors better than posterior confidence by at least 0.05 AUC. **H4** (conformal robustness): 90% coverage falls below 0.85 under at least one shift. H3 and H4 are deliberately falsifiable; a negative result is a finding.

Table 2: Source holdout, $n = 100$ (survey $n = 73$). The calibration layer is fitted on train+dev surveys only. QWK and ECE are evaluated against the generator's latent label and, separately, against the simulated survey.

Metric	vs. latent	vs. survey
QWK	0.718	0.502
Mean absolute error	0.625	0.835
ECE	0.050	0.128
Brier score	0.625	0.737
Calibration slope	0.985	0.688
Large-error rate	0.220	0.384
Conformal 90% coverage	0.976	—

Bootstrap 95% CI for source QWK vs. latent: $[0.605, 0.802]$. The source conformal coverage is above nominal because the base model is refit after the out-of-fold threshold; coverage is therefore conservative on source.

Because the targets contain 100 calls, ECE intervals are wide. We therefore claim H1 only where the paired ΔQWK interval also excludes zero; otherwise the result is reported as directional. This is a power constraint, not a post-hoc rescue.

3.6 Direct LLM pilot

To probe whether the evidence chain is the only way to score, we run a direct scorer: a local `Qwen2.5-3B-Instruct-4bit` model (Qwen Team, 2024) via MLX, temperature 0, on 40 source-holdout calls and their paraphrase twins. The prompt asks for a 1–5 score, a five-class distribution, verbalised confidence, cited turns, an exact quote, and a one-sentence rationale. No calibration layer is fitted for this arm; the question is whether its raw scores and citations are usable. This is a feasibility probe, not a frontier-model benchmark.

4 Preliminary results

4.1 Baseline and the survey ceiling

Table 2 reports the sealed source holdout. Against the latent construct, the chain reaches QWK 0.718 and ECE 0.050, with a calibration slope close to one (0.985). Against the noisy survey that the calibrator was actually fitted to, QWK falls to 0.502, ECE rises to 0.128, and the slope is 0.688. The large-error rate is 0.220 against latent and 0.384 against survey.

This is the first audit finding. A service manager reading a survey-calibrated dashboard would see a system that fits surveys reasonably; a researcher asking whether it measures satisfaction would see a 0.216 QWK gap and a calibration slope of 0.688. Survey anchoring is not construct validation.

4.2 Shift profile

Table 3 and Figure 2 give the main profile. Lexical paraphrase is catastrophic: QWK 0.098, ECE 0.158, slope 1.341, large-error rate 0.440, conformal coverage 0.760. The combined shift is worse still: QWK 0.029, ECE 0.188, slope 0.019, large-error rate 0.590, coverage 0.750. Disfluency and emotional arc are milder (QWK 0.661 and 0.697; ECE 0.071 and 0.065), but their paired ΔQWK intervals include zero: $+0.057\,[-0.022, 0.149]$ and $+0.021\,[-0.015, 0.062]$. Only paraphrase ($+0.619\,[0.474, 0.738]$) and combined ($+0.689\,[0.570, 0.789]$) exclude zero. Persona

Table 3: Shift profile on the sealed source holdout and paired targets. QWK is against latent satisfaction; ECE uses ordinal probabilities; coverage is 90% split conformal. Brackets are 95% bootstrap intervals for QWK; all intervals are wide at $n = 100$.

Condition	QWK	95% CI	ECE	Slope	Large err.	Coverage
Source	0.718	[0.605, 0.802]	0.050	0.985	0.220	0.976
Disfluency	0.661	[0.543, 0.754]	0.071	1.040	0.280	0.980
Emotional arc	0.697	[0.582, 0.783]	0.065	1.018	0.270	0.970
Paraphrase	0.098	[0.025, 0.191]	0.158	1.341	0.440	0.760
Combined	0.029	[−0.014, 0.082]	0.188	0.019	0.590	0.750
Persona mix	0.782	[0.693, 0.848]	0.063	0.979	0.270	0.980

ECE 95% CIs overlap across source and the mild shifts (source $[0.031, 0.164]$, disfluency $[0.026, 0.164]$, emotional arc $[0.032, 0.168]$). The point estimates move in the expected direction, but ECE alone does not separate the mild shifts from source at this sample size.

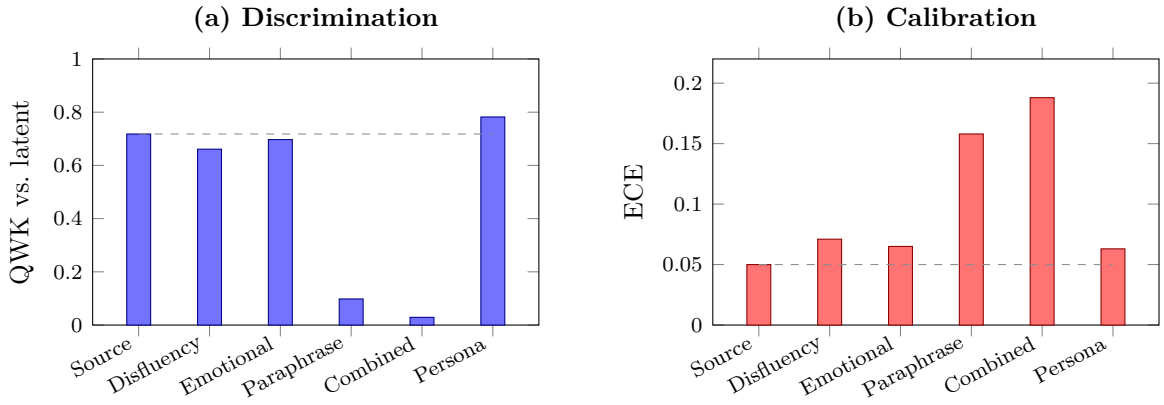


Figure 2: Shift profile. The dashed line marks the source value. QWK collapses under paraphrase and combined shift; ECE rises, but its intervals overlap across the mild shifts (Table 3). The conformal coverage failure is sharper than the ECE change (Section 4.3).

mix is not harmful on this draw (QWK 0.782), but it changes base rates and should not be read as evidence of robustness.

4.3 Where the chain breaks

Table 4 localises the failure. Under paraphrase, the gold evidence-turn hit rate falls from 0.848 to 0.082, the mean number of fact errors rises from 0.51 to 2.61, and quote support falls to 0.260. Resolution accuracy is 0.190 and sentiment accuracy 0.360. The combined shift is only slightly better on facts (2.54 errors) and quote support (0.690). Disfluency degrades resolution accuracy to 0.650 and the gold-turn hit to 0.748 while sentiment accuracy stays at 0.940. Emotional arc leaves resolution accuracy at 0.790 but drops sentiment accuracy to 0.620 and produces a sentiment/resolution conflict on 38% of calls.

The paired consistency results sharpen the claim. Under paraphrase, the average call changes 2.15 facts, 75% of calls move the raw score by more than 0.5, 76% change the predicted label, and the evidence-turn Jaccard is 0.065—the cited turns are almost entirely different even though the latent label is identical. Under disfluency the corresponding numbers are 0.43, 14%, 23%, and 0.915; under emotional arc they are 0.48, 0%, 17%, and 0.970. The lexical shift is therefore a span-layer failure, not a gradual degradation of the same evidence.

The oracle repair confirms the localisation. A broad lexicon that adds the paraphrase synonyms raises paraphrase QWK from 0.098 to 0.718 and the gold-turn hit from 0.082 to 0.848, with no

Table 4: Evidence-chain localisation. "Gold turn hit" is the fraction of facts whose cited turn overlaps a generator cue turn. "Quote supported" checks that the extractor's quote appears in the transcript. Both are oracle-assisted diagnostics, not deployable runtime flags.

Condition	Fact err.	Res. acc.	Sent. acc.	Gold hit	Quote sup.
Source	0.51	0.790	0.940	0.848	1.000
Disfluency	0.91	0.650	0.940	0.748	1.000
Emotional arc	0.83	0.790	0.620	0.867	1.000
Paraphrase	2.61	0.190	0.360	0.082	0.260
Combined	2.54	0.170	0.480	0.247	0.690
Persona mix	0.48	0.810	0.920	0.833	0.990

Fact error is the mean number of incorrect facts (of four) per call. Oracle-assisted columns use the generator's gold cue turns; the symbolic extractor is a lower bound, not an LLM.

source change (0.718 to 0.718). But the repair was written by looking at the paraphrase list, so it is an *oracle upper bound*, not a deployable fix. Its value is diagnostic: if a known synonym list recovers the chain, the failure is lexical, not a fundamental limit of fact-based scoring.

4.4 Calibration is not the bottleneck, but it is not enough

The calibration layer behaves as designed on source: the ordinal model's temperature is 1.078, the conformal threshold is 0.893, and the reliability curve is close to the diagonal against latent. Under paraphrase, post-hoc isotonic calibration recovers QWK only to 0.107; the problem is that the facts entering the calibrator are wrong or missing. Under combined shift the calibration slope is 0.019—the score carries almost no signal about the latent construct. Calibration cannot repair an evidence chain that has already broken upstream.

The survey-versus-latent gap is a second calibration lesson. A survey-anchored layer can improve fit to the survey (isotonic QWK 0.549 vs. 0.502 raw on survey) while moving away from the construct (isotonic QWK 0.717 vs. 0.718 raw on latent). The layer is doing what it was asked to do; the question is whether the survey is the right target.

4.5 Audit signals and the failed deferral policy

Table 5 compares posterior confidence with three label-free flags for predicting large errors. Posterior confidence is at or below chance for ranking large errors on source (0.372), disfluency (0.275), and emotional arc (0.354). The missing-resolution flag beats confidence on source (0.589 vs. 0.372), disfluency (0.624 vs. 0.275), and emotional arc (0.648 vs. 0.354), but not on paraphrase (0.534 vs. 0.478) or combined (0.520 vs. 0.567). The conflict flag is weak everywhere; the low-confidence flag is weak on source and paraphrase.

The deferral test is the decision-relevant check. A gate combining confidence with the three flags improved the large-error rate at 80% coverage by 0.0 pp (source), −2.5 pp (disfluency), and −1.2 pp (emotional arc), but worsened paraphrase by +2.5 pp, and did not improve AUC (max ΔAUC +0.015). We discard it. The interpretation is that "no evidence" is often the scorer's honest default, not the dangerous case; the dangerous case is *confident wrong evidence*. A presence flag cannot see that. A contradiction audit—does the cited span entail the typed fact?—is the next candidate.

4.6 Robustness versus auditability: the direct LLM contrast

The direct 3B scorer is the most informative contrast in the pilot. On the same 40 source calls and their paraphrase twins, the symbolic chain falls from QWK 0.746 to 0.113 under paraphrase,

Table 5: AUC for predicting a large error ($|\hat{y} - y| > 1$) on latent satisfaction. Confidence is the ordinal posterior maximum. Flags are label-free; "missing res." is no cited resolution evidence, "conflict" is a sentiment/resolution contradiction, "low conf." is mean fact confidence below 0.55.

Condition	Confidence	Missing res.	Conflict	Low conf.
Source	0.372	0.589	0.467	0.522
Disfluency	0.275	0.624	0.477	0.602
Emotional arc	0.354	0.648	0.426	0.525
Paraphrase	0.478	0.534	—	0.494
Combined	0.567	0.520	—	0.483
Persona mix	0.439	0.524	0.557	0.506

AUC is a ranking diagnostic, not a deferral policy. The conflict flag is undefined when no sentiment/resolution pair is extracted. These AUCs are exploratory and have no interval yet.

Table 6: Direct 3B scorer versus the symbolic evidence chain on the same 40 source-holdout calls and their paraphrase twins. "Large error" is $|\hat{y} - y| > 1$; ECE is five-class probability ECE for the direct arm and ordinal ECE for the symbolic arm.

Arm	Source QWK	Para. QWK	Source large	Para. large	Source ECE	Para. ECE
Symbolic chain	0.746	0.113	0.200	0.450	0.090	0.199
Direct 3B LLM	0.669	0.667	0.150	0.150	0.575	0.593

Direct-arm audit: confidence ECE 0.498/0.478; mean confidence 0.874/0.854; citation precision 0.604/0.533; quote support 0.95/0.90; parse rate 1.000/0.975. One local 3B model, 40 calls per condition, not a frontier-model result.

while the direct LLM holds QWK at 0.669 to 0.667 (Table 6, Figure 3). It also has a lower large-error rate (0.150 vs. 0.450 under paraphrase). But its five-class probability ECE is 0.575 on source and 0.593 on paraphrase, its verbalised confidence averages 0.874 and 0.854, and its citation precision is 0.604 and 0.533. Its quotes are supported 95% and 90% of the time, and cited turns hit a gold cue turn on 97.5% and 92.5% of calls, but many of those turns are not the evidence for the score. The model is robust to vocabulary and weakly auditable; the symbolic chain is auditable and brittle.

4.7 What the preliminary evidence supports

Table 7 records the keep/discard decisions. Two are clear: broadening the lexicon is a diagnostic upper bound, not a deployable fix; isotonic and ordinal calibration serve different jobs and should both be kept. The runtime gate is discarded. The direct LLM is kept as a pilot only.

The report's working claims, stated with the caveats they require, are:

1. **Lexical shift is an extractor-level failure, not a calibration-layer failure.** The symbolic chain loses its spans (gold hit 0.848 → 0.082), post-hoc isotonic recovers QWK only to 0.107, and the direct LLM holds QWK at 0.667. The fragile interface is span/fact extraction.

2. **ECE is too blunt at this sample size.** Its point estimates rise in the expected direction, but the source interval $[0.031, 0.164]$ overlaps the mild-shift and paraphrase intervals. Conformal coverage is sharper: 0.976 on source, 0.760 under paraphrase, 0.750 under combined. An audit should report discrimination, evidence recovery, and coverage together.

3. **Survey anchoring is not construct validation.** The source chain reaches QWK 0.718 against latent but 0.502 against survey; ECE 0.050 vs. 0.128; slope 0.985 vs. 0.688.

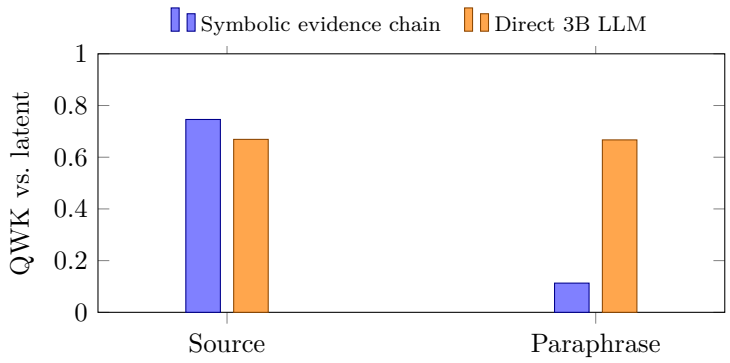


Figure 3: The robustness–auditability contrast. The symbolic chain is more accurate on source and collapses under paraphrase; the direct LLM is less accurate on source, stable under paraphrase, and badly calibrated. The comparison uses the same 40 calls.

Table 7: Keep/discard ledger for the preliminary cycle.

Card	Change	Decision
C0	Strict lexicon + source calibration (baseline)	Reference
C1	Broad lexicon with paraphrase synonyms	Keep (diagnostic, circular)
C2	Isotonic point calibration alongside ordinal probabilities	Keep both
C3	Runtime evidence gate for abstention	Discard
C4	Direct small-LLM scoring pilot	Pilot only

4. **Evidence-presence flags are diagnostic, not a deferral policy.** Missing-resolution AUC beats confidence on source, disfluency, and emotional arc, but the gate improves selective risk in only two of four conditions and worsens paraphrase. The next audit should test contradiction, not presence.

5. **Robustness and auditability trade off in this pilot.** The direct LLM is robust to paraphrase but overconfident and weakly traceable; the symbolic chain is traceable but brittle. The next design should put a robust extractor inside the auditable chain.

5 Discussion

The preliminary results suggest three shifts in how satisfaction scoring should be evaluated.

From score accuracy to evidence consistency. A QWK against a survey is a property of the score, not of the measurement chain. Two systems can have the same QWK while one is supported by the right spans and the other by plausible-sounding but wrong evidence. The paired design makes this visible: under paraphrase, 76% of predicted labels change even though the latent label is fixed, and the cited evidence turns are almost entirely different (Jaccard 0.065). An audit that samples only scores will miss this; an audit that samples spans will catch it.

From calibration error to a calibration profile. ECE is a useful summary but a poor sole trigger. In our pilot, ECE moves in the expected direction under shift, yet its intervals overlap and it does not distinguish the mild shifts from source. Conformal coverage, calibration slope, and the large-error rate give a fuller picture. The broader lesson is consistent with the calibration-cliff literature (Wang et al., 2026): post-hoc calibration is local to the distribution it was fitted on, and a single number can hide the failure.

From presence to contradiction. The failed gate is informative. Missing evidence is often the scorer's honest default, and therefore not a strong error signal. Confident wrong evidence is the dangerous case, and a presence flag cannot detect it. A contradiction audit—checking whether the cited span entails the typed fact—is a natural next step, and it aligns with evidence-boundary designs in customer-service agents (Wu et al., 2026).

Implications for service governance. For ECIS Track 10, the practical implication is that synthetic-log-based satisfaction dashboards need *shift-aware* audits before they are used for coaching, compliance, or resource allocation. The audit should be part of the measurement instrument, not an afterthought: which spans support the score, does the score survive a meaning-preserving rewrite, and does the calibration still cover under a plausible synthetic shift? The answers are governance-relevant because they determine whether a manager is looking at satisfaction or at a survey-fitting artefact.

6 Threats to validity

Synthetic ground truth. The latent label is a structural equation, not a human judgment. The survey is a noisy draw from it, and the annotators are simulated. Every calibration number inherits that construction. The design is useful for localising pipeline failures, not for estimating real-world CSAT accuracy.

Small targets and wide intervals. One hundred calls per shift gives QWK 95% CIs of roughly ± 0.06–0.10. Only paraphrase and combined have paired ΔQWK intervals excluding zero. The disfluency and emotional-arc effects are directional, not established. The next run should size targets for a paired effect or report the mild shifts as inconclusive.

The offline extractor is a lower bound. It was written against the templates, so its source fact accuracy is optimistic and its paraphrase collapse is partly a property of the lexicon. The broad-lexicon repair is circular by construction. The direct LLM partially addresses this, but it is a scorer, not an extractor: the evidence chain with an LLM extractor remains unrun.

The LLM pilot is small. One 3B open model, 40 calls per condition, a single prompt, temperature 0. It cannot rank models or support a claim about GPT-class systems. Its value is to show that the audit instrument transfers and that verbalised confidence is badly overconfident in this setting.

Synthetic transformations. The shifts operationalise Mirage-style dimensions, but they do not reproduce real disfluency, accents, code-switching, or the full texture of emotional expression. The persona-mix target is unpaired and changes base rates; its high QWK should not be read as robustness.

Calibration implementation. The conformal sets over-cover on source (0.976 vs. nominal 0.90) because the base model is refit after the out-of-fold threshold. Coverage failures under shift are still informative, but the source baseline is conservative. The 0.5 raw-score tolerance is coarse for sentiment-only changes; label-change and expected-score deltas are therefore reported alongside it.

7 Conclusion and next steps

This report started a shift-aware audit of LLM-based satisfaction scoring on synthetic call logs. The preliminary evidence is consistent with a simple claim: the calibration layer is not the main source of fragility; the span/fact interface is. A known synonym list restores the symbolic chain, and a direct 3B LLM is robust to the same paraphrase, but the direct model is overconfident and only weakly traceable. Robustness and auditability trade off, and the next design should combine them.

The immediate next experiments are:

1. Run a frontier LLM extractor through the typed-fact prompt on the same 100-call holdouts, then score deterministically. Test whether it keeps the direct arm's paraphrase robustness while retaining spans.
2. Add a contradiction audit: for each cited span, ask whether it entails the typed fact (NLI or a second judge), and test whether contradiction rate predicts large errors better than presence flags.
3. Run a frontier direct scorer with proper probability elicitation (verbalised histograms or token logprobs) and temperature scaling on a source dev split.
4. Add a human-adjudicated subset (30–50 calls) to test whether the survey ceiling is an artefact of the simulated annotators.
5. Pre-register the tolerance and the minimum detectable effect; the current 100-call targets are underpowered for the mild shifts.

Data and code. All data are synthetic and generated by the accompanying harness. The notebooks reproduce the baseline, the shift profile, the audit, and the keep/discard ledger. The LLM pilot is optional and requires the local model runtime.

References

Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification, 2021. arXiv:2107.07511.

call-eval maintainers. call-eval: Post-call customer experience evaluation. https://github.com/henryzhangpku/call-eval, 2026. Repository returned 404 on 20 October 2026; architecture documented in the archived project guide (JevList, 2026).

Alessandro Chincha, Cristian Bonifacio, and Cesar Salas. BellTeus: Call center performance evaluation system using the large language model. In *2025 11th International Conference on Computer and Communications (ICCC)*, pages 472–476, 2025. doi: 10.1109/ICCC68654.2025.11438128.

Jacob Cohen. Weighted kappa: Nominal scale agreement provision for scaled disagreement or partial credit. *Psychological Bulletin*, 70(4):213–220, 1968. doi: 10.1037/h0026256.

Rishikesh Devanathan, Varun Nathan, and Ayush Kumar. Mirage: A diagnostic framework for evaluating the realism of synthetic contact center dialogue generation. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 25187–25227, 2026. doi: 10.18653/v1/2026.findings-acl.1261.

ECIS. Track 10: Design and management of services in the age of AI. https://ecis2027.uni.lu/track-10/, 2027. Accessed 20 October 2026.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, 2017.

JevList. call-eval — Jev project guide. https://jevlist.ai/projects/call-eval, 2026. Accessed 20 October 2026; reviewed upstream at commit abba9792be56.

Natalia Krawczyk, Alicja Kasicka, Bartosz Przybył, and Justyna Gromada. Interpretable satisfaction modeling in sales dialogues: A metrics-based approach validated against text-based model. In *Proceedings of the 27th Annual Meeting of the Special Interest Group on Discourse and Dialogue*, pages 565–584, 2026. doi: 10.18653/v1/2026.sigdial-1.40.

Aakriti Kumar, Nalin Poungpeth, Diyi Yang, Erina Farrell, Bruce L. Lambert, and Matthew Groh. When large language models are reliable for judging empathic communication. *Nature Machine Intelligence*, 8:173–185, 2026. doi: 10.1038/s42256-025-01169-6.

Etienne Manderscheid and Matthias Lee. Predicting customer satisfaction by replicating the survey response distribution, 2024. arXiv:2411.12539.

Alexandru Niculescu-Mizil and Rich Caruana. Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning*, pages 625–632, 2005. doi: 10.1145/1102351.1102430.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

Qwen Team. Qwen2.5 technical report, 2024. arXiv:2412.15115.

Heydar Soudani, Roxana Petcu, Evangelos Kanoulas, and Faegheh Hasibi. A survey on recent advances in conversational data generation. *ACM Computing Surveys*, 58:1–37, 2026. doi: 10.1145/3795686.

Zhengmeng Wang, See Kee Ng, Tinglei Liu, and Suriati Binti Saad. CARE: A calibration-aware framework for cyberbullying detection under distribution shift. *IEEE Access*, 14:123260–123284, 2026. doi: 10.1109/ACCESS.2026.3721881.

Chunming Wu, Dafei Qiu, Congde Yuan, Charles Quan, Jun Wu, Suipeng Li, Mo Wu, Gavin Xie, Hope Chen, and Max Yao. Evidence-in-the-loop: Trace-driven optimization for customer-service LLM agents, 2026. arXiv:2607.18039.

Jie Zhu, Huaixia Dou, Junhui Li, Lifan Guo, Feng Chen, Chi Zhang, and Fang Kong. Evaluating, synthesizing, and enhancing for customer support conversation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 35185–35194, 2026. doi: 10.1609/aaai.v40i41.40825.